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Jewelry cleaning device

Abstract

A jewelry cleaning device includes a housing, a cleaning mechanism, and a ring holder. The ring holder can be moveable with respect to the housing among at least two positions. The ring holder can include a moveable grip. The housing can include an upper housing assembly and a lower housing assembly. The upper housing assembly can include a lid moveable with respect to an upper housing between an open position and a closed position. The upper housing assembly can move with respect to the lower housing assembly between an open position and a closed position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority benefit of U.S. Provisional Patent Application No. 63/275,014 filed on Nov. 3, 2021, entitled “Jewelry Cleaning Device,” which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

(1) Jewelry cleaning devices exist that include a bath of cleaning solution where the bath vibrates in the sonic or ultrasonic range to dislodge particles from the jewelry. Drawbacks of these known jewelry cleaning devices include cumbersome ways in which the article of jewelry is held in the jewelry cleaning device during a cleaning cycle. What is needed is an improved jewelry cleaning device.

SUMMARY OF THE INVENTION

(2) In view of the foregoing, an example of a jewelry cleaning device includes a housing, a bath located in the housing, a cleaning brush mounted to the housing, and a ring holder connected with the housing. The bath is configured to hold a cleaning solution. The ring holder is configured to hold a ring to be cleaned by the jewelry cleaning device. The ring holder is moveable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath.

(3) In another aspect, the ring holder is connected with or part of an arm that pivots with respect to the housing.

(4) In another aspect, at least a portion of the cleaning brush is disposed in the cleaning solution in the bath.

(5) In another aspect, a transducer cooperates with the bath for vibrating the bath.

(6) Another aspect of the invention comprises a plurality of bristles connected with the housing and positioned above the cleaning brush with respect to a support surface on which the housing rests. The ring holder positions the ring above the plurality of bristles when in the first position and the ring holder positions the ring below the plurality of bristles when in the second position.

(7) In yet another aspect, the cleaning brush further comprises a cleaning brush housing with bristles and a vibration motor disposed thereon.

(8) In another aspect, the ring holder includes a fixed grip and a movable grip that is moveable with respect to the fixed grip.

(9) Another aspect of the invention includes an upper housing assembly and a lower housing assembly. The upper housing assembly includes a lid pivotable with respect to an upper housing between an open position and a closed position.

(10) In yet another aspect, the ring holder comprises a pair of opposed members spring biased in an open position. The opposed members are configured to fit inside a band portion of the ring to hold the ring in position.

(11) In another aspect of the invention a silicone material is disposed on at least one of the opposed members.

(12) Another aspect of the invention comprises one or more magnets disposed on one of the upper housing assembly and the lower housing assembly. The magnets are configured to provide a force holding the upper housing assembly and the lower housing assembly together.

(13) In yet another aspect, the jewelry cleaning device further comprises a raised portion on one of the upper housing assembly and lower housing assembly and an opening on the other of the upper housing assembly and lower housing assembly such that the raised portion provides an alignment guide for connecting the upper housing assembly to the lower housing assembly.

(14) In another example of a jewelry cleaning device, such a jewelry cleaning device includes a housing. A bath is located in the housing and configured to hold the cleaning solution. A cleaning brush housing is mounted to the housing. The cleaning brush housing has bristles and a vibration motor disposed thereon. A ring holder is connected with the housing and configured to hold the ring.

- (15) The ring holder is movable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath.
- (16) A transducer cooperates with the bath for vibrating the bath.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a jewelry cleaning device with a lid in an open position.
- (2) FIG. 2 is a perspective view of the jewelry cleaning device with the upper housing assembly shown in an open position.
- (3) FIG. 3 is a cross-sectional view taken through line 3-3 in FIG. 1, but with the lid shown in the closed position.
- (4) FIG. 4 is a cross-sectional view taken through line 4-4 in FIG. 1 with the lid in the closed position.
- (5) FIG. 5 is a perspective view of the jewelry cleaning device with the lid, an upper housing and upper brush removed.
- (6) FIG. 6 is a schematic cross-sectional view of the jewelry cleaning device with a ring holder in a first position.
- (7) FIG. 7 is a schematic cross-sectional view of the jewelry cleaning device with the ring holder in a second position.
- (8) FIG. 8 is a schematic cross-sectional view of the jewelry cleaning device with the ring holder in a third position.
- (9) FIG. 9 is a perspective view of an alternate embodiment of the jewelry cleaning device of the present invention.
- (10) FIG. 10 is a bottom perspective view of the jewelry cleaning device shown in FIG. 9.
- (11) FIG. 11 is an exploded perspective view of the jewelry cleaning device shown in FIG. 9.
- (12) FIG. 12 is a partial perspective view of a portion of an upper assembly of the jewelry cleaning device of FIG. 9.
- (13) FIG. 13A is a side elevational view of a portion of the ring holder mechanism of the jewelry cleaning device of FIG. 9.
- (14) FIG. 13B is a partial detailed side elevational view showing a ring on the opposed members.
- (15) FIG. 14 is a top plan view of the lower assembly of the jewelry cleaning device of FIG. 9.
- (16) FIG. 15 is a perspective view of the lower assembly of the jewelry cleaning device of FIG. 9.
- (17) FIG. 16 is a front elevational view of the jewelry cleaning device of FIG. 9.
- (18) FIG. 17 is a sectional view taken along lines 17-17 of FIG. 16.

DETAILED DESCRIPTION

- (19) With reference to FIG. 1, a jewelry cleaning device 10 is shown generally including a housing, which in the illustrated embodiment includes an upper housing assembly 12 and a lower housing assembly 14. The jewelry cleaning device 10 further includes a cleaning mechanism that, with reference to FIG. 2, can include cleaning brushes 16 and a bath 18 housed within the housing. With reference back to FIG. 1, the jewelry cleaning device 10 further includes a ring holder 20 configured to hold a ring 22 to be cleaned by the jewelry cleaning device 10.
- (20) The upper housing assembly 12 generally includes an upper housing 30, which in the depicted embodiment is in the shape of a frustum at an upper portion with vertical sidewall sections 28 (FIG. 3) beneath the frustum shaped portion. The upper housing 30 includes an upper opening 32 surrounded by a recess 34 formed in a horizontally oriented top wall section 36. A lid 38 connects with the upper housing 30 at a hinge connection 42, which allows the lid 38 to rotate about a lid pivot axis 44 with respect to the upper housing 30. Alternatively, the lid 38 could be moveable with

respect to the upper housing **30** in another manner, e.g., being simply removable from the upper housing **30**. In the illustrated embodiment, the lid **38** is made from a transparent material and is generally pyramid-shaped. With reference to FIG. **3**, the upper housing **30** includes standoffs **46** to facilitate connection between the upper housing **30** and an upper housing insert **50**. The upper housing insert **50** includes an upper horizontal portion **52** that defines an upper insert opening **54**. The upper housing insert **50** also includes downwardly depending vertical sections **56** that are coplanar with the lower horizontal sections of the upper housing **30**.

(21) With reference to FIG. **3**, the lower housing assembly **14** includes a lower housing **80** connected with a lower housing lid **82**. The lower housing **80** includes a horizontally oriented base **84**. A rubber-like (or similar non-slip material) mat **86** can attach to the base **84**. The lower housing **80** can include vertical sidewall sections **88** to define a hollow interior that receives the lower housing lid **82**. Similar to the lower housing **80**, the lower housing lid **82** is generally basin-shaped to define a bath assembly cavity **92**. The lower housing lid **82** also includes a hinge portion **94** (see FIG. **2**) for connecting with an associated hinge portion **96** (see FIG. **4**) provided on the upper housing **30** to pivotally attach the upper housing assembly **12** to the lower housing assembly **14**. With reference to FIG. **2**, the upper housing assembly **12** is pivotable about a housing pivot axis **98** with respect to the lower housing assembly **14** between an open position (shown in FIG. **2**) and a closed position (shown in FIG. **1**).

(22) With reference to FIG. **2**, with the upper housing assembly **12** in the open position, a bath assembly **110**, which is received in the lower housing assembly **14**, is accessible for removal from the lower housing assembly **14**. The bath assembly **110** can be removed from the lower housing assembly **114** for cleaning and maintenance reasons. Also, the bath assembly may be removed to allow for dumping and refilling the cleaning solution that is to be used in the jewelry cleaning device **10**.

(23) The bath assembly **110** generally includes a bath lower housing **120** that is received in the bath assembly cavity **92** formed in the lower housing lid **82**. The bath lower housing **120** is similar in configuration to the lower housing lid **82** in that it is generally basin-shaped having a horizontal base with vertical side sections. The bath lower housing **120** has an open top to define a bath cavity **122** in which is received a bath **124** that is configured to hold the cleaning solution typically used to clean jewelry. The bath **124** defines a fill line **126** (only shown in FIG. **3**), which is a line to which the cleaning solution received in the bath **124** is to be filled. The bath assembly **110** further includes a bath upper housing **128** that connects with an upper edge of the bath **124** and sandwiches a bath seal **132** between the bath upper housing **128** and the upper edge of the bath **124**. The bath assembly **110** further includes a transducer **134** mounted to the bath **124**. The transducer **134** electrically connects with a first electrical connector **136** (depicted schematically in FIG. **4**), which can be provided on a lower surface of the bath lower housing **120**. A second electrical connector **138** (depicted schematically in FIG. **4**), which connects with a power source (not shown) via an electrical connection line **142** (depicted schematically in FIG. **4**), is provided on the lower surface of the bath lower housing **120** within the bath assembly cavity **92**. The transducer **134** is provided for vibrating the bath **124** in the ultrasonic range when turned ON to receive electrical power. The first electrical connector **136** is disconnectable with the second electrical connector **138** when the bath assembly **110** is removed from the lower housing assembly **14**. Similarly, when the bath assembly **110** is placed back into the lower housing assembly **14**, the first electrical connector **136** reconnects with the second electrical connector **138** so that power can be provided to the transducer **134**.

(24) As mentioned above, the jewelry cleaning device **10** includes cleaning brushes **16**, which can be part of cleaning brush assemblies **160**, **162** mounted to the housing. With reference to FIG. **4**, in the illustrated embodiment the cleaning brush assemblies **160**, **162** are mounted to the upper housing assembly **12**, and more particularly to the upper housing insert **50**. The cleaning brush assemblies **160**, **162** are opposed from one another and each is similar in configuration; therefore,

the first cleaning brush assembly **160** will be described with particularity with the understanding that the second cleaning brush assembly **162** is similarly configured.

(25) With reference also to FIG. 5, the first cleaning brush assembly **160** includes a front brush housing **166** connected with a rear brush housing **168**. The rear brush housing **168** includes a wire routing opening **172** (visible in FIG. 5). With reference to FIG. 4, a vibration motor **176** is positioned between the front brush housing **166** and the rear brush housing **168** and receives power through wiring **178** (only a portion shown in FIG. 5) passed through the wire routing opening **172**. With reference to FIG. 4, the first cleaning brush assembly **160** includes a bristle mounting pad **180** that mounts to the front brush housing **166**. A plurality of bristles **182** extend from the bristle mounting pad **180** away from the front brush housing **166** toward the second cleaning brush assembly **162**. The bristle mounting pad **180** can releasably connect to the front brush housing **166** by being slid into the front brush housing **166** and snap into place so that the bristle mounting pad **180** with the associated plurality of bristles **182** can be replaceable. As evident in FIG. 3, at least a portion of each of the cleaning brush assemblies **160**, **162** can be disposed below the fill line **126** of the bath **124**. With reference to FIG. 5, a mounting bracket **186** connects the front brush housing **166** and the rear brush housing **168** to the upper housing insert **50** via fasteners **188**. Vibration isolators **192**, which can be made from a rubber-like material, connect the rear brush housing **168**, which is connected with the front brush housing **166**, to the mounting bracket **186**. As such, when the vibration motor **176** is operating, the first cleaning brush assembly **160** vibrates and the vibration isolators **192** dampen the vibration with respect to the mounting bracket **186**.

(26) With reference to FIGS. 1 and 5, the ring holder **20** is part of a ring holder mechanism **202**. With particular reference to FIG. 5, the ring holder mechanism **202** includes a ring arm motor **204** operatively connected with a ring arm **206** that includes an axle section **208** and an arm extension **212** extending from the axle section **208**. The axle section **208** connects with an output shaft (not visible) of the ring arm motor **204** for rotational movement about a ring arm axis **214**. With reference to FIG. 4, the ring holder mechanism **202** further includes a ring mount front **216** connected with a ring mount back **218** that each extend upwardly from the arm extension **212** (see FIG. 5) of the ring arm **206**. The ring holder **20** includes at least two grips, which in the illustrated embodiment include a fixed grip **222** and a moveable grip **224**. With reference to FIG. 3, the fixed grip **222** is positioned above the moveable grip **224** with respect to a support surface on which the housing **12**, **14** rests. In the illustrated embodiment, the fixed grip **222** is curved to generally follow a circle so that the two grips **222**, **224** define at least three contact points on the circle **226** which can be an inner circumference of the associated ring **22** to be cleaned by the jewelry cleaning device **10**. With continued reference to FIG. 3, a first contact point **228** is located on the circle **226**. A second contact point **232** is also located on the circle **226** and is disposed on a first side of a diameter **234** intersecting the first contact point **228**. A third contact point **236** is also located on the circle **226** but is disposed on a second, opposite side of the diameter **234** intersecting the first contact point **228**. In the illustrated embodiment, one fixed grip **222** is provided on the ring holder **20**; however, two fixed grips each associated with the second contact point **232** and the third contact point **236**, respectively, can be provided. Both of the grips **222** and **224** can be covered by an elastomer or other rubber-like material. With reference to FIG. 3, the second contact point **232** is angularly offset from the third contact point **236** about 45°. If desired, the angular offset can be reduced to at least about 15°. This angular offset provides a more robust connection for the associated ring **22** to be cleaned during the cleaning process.

(27) With reference back to FIG. 4, a spring **242** is trapped between the ring mount front **216** and the ring mount back **218**. The ring mount back **218** includes a spring seat **244** and the moveable grip **224** also includes a spring seat **246** such that the spring **242** is trapped between the spring seat **244** on the ring mount back **218** and the spring seat **246** on the moveable grip **224**. The spring **242** is a compression spring that biases the moveable grip **224** downward per the orientation shown in FIG. 4. The moveable grip **224** includes a finger **250** that extends through a slot **252** provided in the

ring mount front **216**. The moveable grip **224** is moveable with respect to the fixed grip **222** and biased by the spring **242** to accommodate rings of different sizes.

(28) As mentioned above, the ring arm motor **204** drives the ring arm **206** for rotation about the ring arm axis **214**. With reference to FIGS. **6**, **7**, and **8** the ring holder **20** is moveable with respect to the housing **12**, **14** among at least three positions. FIG. **6** depicts a first position where the ring holder **20** positions the associated ring **22** offset from the bath **124** and the cleaning brush assemblies **160**, **162**. FIG. **7** depicts a second position where the ring holder **20** positions the associated ring **22** with respect to the cleaning brush assemblies **160**, **162** for contact with the cleaning brushes. FIG. **8** depicts a third position where the ring holder **20** positions the associated ring **22** in the bath **124** beneath the fill line **126**.

(29) With reference back to FIG. **5**, a switch assembly, which can include a first limit switch **260** and a second limit switch **262** can control the ring arm motor **204** and thus the position of the ring holder **20**. For example, the ring **22** to be cleaned can be mounted to the ring holder **20** in the manner shown in FIG. **1**. The lid **38** can then be placed into the closed position shown in FIG. **4**. The jewelry cleaning device **10** can then be turned on a manner that will be described in more detail below. With the jewelry cleaning device **10** in the ON state, power is delivered to the transducer **134** and the vibration motors **176**. The ring arm **206** then moves from the position shown in FIG. **6** to the lower cleaning position shown in FIG. **8**. The angular displacement between the position shown in FIG. **6** and the position shown in FIG. **8** is about 80° (plus or minus 5°). The angular displacement of about 80° trips the first limit switch **260** (FIG. **5**) and the ring arm **206** pauses in the position shown in FIG. **8** for a soak time, which can be a predetermined amount of time such as about 20 seconds. After the predetermined amount of time has elapsed, the ring arm **206** moves to the position shown in FIG. **7**, which can be about 45° angularly offset from the position shown in FIG. **6** (plus or minus about 5°). With the ring arm **206** shown in the position in FIG. **7**, the second limit switch **262** trips and the ring arm **206** can return to the position shown in FIG. **8**. Movement between the positions shown in FIGS. **7** and **8** can be repeated for a predetermined number of cycle times, such as five cycle times. After five cycles, for example, between the position shown in FIG. **7** and the position shown in FIG. **8**, the ring arm **206** can remain in the position shown in FIG. **8** for a second predetermined amount of time which can be about 120 seconds. After the second predetermined amount of time elapses, power can be turned OFF to the transducer **134** and the vibration motors **176**. The ring arm **206** can then return to the position shown in FIG. **6**.

(30) With reference back to FIG. **5**, the jewelry cleaning device **10** can further include a lid motor **270** that drives a pinion **272**. The pinion **272** can be operatively connected with a gear (not shown) connected with the lid **38**. After the cleaning cycle described in the aforementioned paragraph, the lid motor **270** can receive power to drive the pinion **272** to open the lid **38**. With reference back to FIG. **1**, the jewelry cleaning device **10** can further include a light source **280** connected with the housing, which in the illustrated embodiment is connected with the upper housing assembly **12** and more particularly is provided in the recess **34** on the upper housing **30**. The light source **280**, which can be an LED, is positioned on the housing and configured to direct light toward the ring holder **20** when the ring holder **20** is in the first position, which is shown in FIGS. **1** and **6**. After the cleaning cycle described above power can be delivered to the light source **280** to illuminate the ring **22**.

(31) With reference to FIGS. **1** and **4**, upper brushes **302**, **304** are positioned above the cleaning brush assemblies **160**, **162**. Each upper brush **302**, **304** is similarly configured. As such, the second upper brush **304** will be described with particularity with the understanding that the first upper brush **302** is similarly configured. The second upper brush includes a brush mount **306** and a plurality of silicone bristles **308** extending from the brush mount **306** toward the first upper brush **302**. The brush mount **306** mounts to the upper housing **30** and the plurality of silicone bristles **308** extend into the upper opening **32** to obscure the internal components of the jewelry cleaning device

10. The plurality of silicone bristles **308** can also operate as a wiper to remove residual cleaning solution from the ring **22** being cleaned by the jewelry cleaning device **10**. With the ring holder **20** in the first position (shown in FIGS. **1** and **6**) the ring holder **20** positions the associated ring **22** above the plurality of silicone bristles **308**. With reference to FIGS. **6** and **7**, the ring holder **20** positions the associated ring **22** below the plurality of bristles **308** when both in the second position (FIG. **7**) and the third position (FIG. **8**).

(32) With reference back to FIG. **2** and also with reference to FIG. **1**, the ring holder **20** and the cleaning brush assemblies **160**, **162** are mounted to the upper housing assembly **12** for movement with the upper housing assembly **12** as the upper housing assembly **12** moves with respect to the lower housing assembly **14**. This allows for easy removal of the bath assembly **110**. With reference to FIG. **2**, a button **320** provided on the lower housing assembly **14** can operatively connect with a latch **322** also on the lower housing assembly **14**. Depressing the button **320** can move the latch **322** into disengagement with a catch **324** provided on the upper housing assembly **12** to allow for movement of the upper housing assembly **12** with respect to the lower housing assembly **14**. The upper housing assembly **12** can pivot about the housing pivot axis **98** with respect to the lower housing assembly **14**. Alternatively, other types of movement of the upper housing assembly **12** with respect to the lower housing assembly **14** could be provided, for example, complete removal of the upper housing assembly **12** from the lower housing assembly **14**. When pivotal movement is desired, and with reference back to FIG. **3**, weights, e.g., lead weights **330**, can be provided in the lower housing assembly **14** to prevent the jewelry cleaning device **10** from falling backwards when the upper housing assembly **12** is pivoted with respect to the lower housing assembly **14**.

(33) With reference back to FIG. **1**, a first button **340** and a second button **342** can be provided on the upper housing assembly **12**. The first button **340** can operate a switch (not shown) to control delivery of power to the lid motor **270** (FIG. **5**) such that operation of the first button **340** results in the lid **38** moving from the closed position toward the open position, which is shown in FIG. **1**. The second button **342** can operate a switch electrically connected with the ring arm motor **204**, the vibration motors **176**, and the transducer **134** such that when one depresses the second button **342** the jewelry cleaning device **10** is turned ON into a cleaning mode. With reference back to FIG. **4**, power can be provided to the jewelry cleaning device **10** via an electrical cord **350** (depicted schematically), which can connect with a wall electrical outlet.

(34) Referring generally to FIGS. **9-17**, and initially to FIG. **9**, an alternate embodiment of the jewelry cleaning device **10** is shown. A jewelry cleaning device **1000** is shown generally including a housing, which in the illustrated embodiment includes an upper housing assembly **1003** and a lower housing assembly **1006**. The upper housing assembly **1003** may be provided with a lid **1009** that may be pivotally attached thereto. The lid **1009** may rotate between an open position shown in FIG. **9** and a closed position shown in FIGS. **16-17**. In the center of the upper housing assembly **1003** a recessed portion **1012** borders an opening **1015**. The upper portion of a ring holder mechanism **1018** extends upward through the opening **1015** into the space defined by the lid **1009**. A ring **1021** is attached to the ring holder mechanism **1018** as will be described in greater detail herein. A plurality of bristles **1024** are shown in the opening **1015** such that the ring **1021** passes through the bristles **1024** when it moves downward as described in greater detail herein. The upper housing assembly **1003** may removably attach to the lower housing assembly **1006**. In the embodiment shown, the upper housing assembly **1003** is lifted vertically to separate the upper housing assembly **1003** from the lower housing assembly **1006**. The upper housing assembly **1003** and the lower housing assembly may have electrical contacts between them configured to make an electrical connection so that power from the lower housing assembly **1006** can be distributed to the motors in the upper housing assembly **1009** such as the ring arm motor **1063** and the vibration motor **1099** in the brush housing **1084** described below.

(35) Turning to FIG. **10**, the bottom surface **1027** of the lower housing assembly **1006** may be provided with ribbed portions **1030** to provide a non-slip surface. The bottom surface **1027** of the

lower housing assembly **1006** may also be provided with one or more vent openings **1033** to provide ventilation and cooling. Vents **1036** may also be provided in an angled outer wall **1039** of the lower housing assembly **1006** for additional ventilation. In addition, the jewelry cleaning device **1000** may also be provided with one or more electric fans **1042** to provide for cooling the motors and electronics contained with the jewelry cleaning device **1000**. A groove **1043** may be formed between the upper housing assembly **1003** and the lower housing assembly **1006** to provide for manually separating the two assemblies.

(36) In FIG. **11**, the upper housing assembly **1003** is separated from the lower housing assembly **1006**. The top surface **1045** of the lower housing assembly **1006** may be provided with a pair of recessed portions **1048**, **1051** that provide channels (best shown in FIGS. **14-15**) for pouring the cleaning solution out of the bath **1052** in the lower housing assembly **1006**. The channels formed by the recessed portions **1048**, **1051** provide a pathway to avoid spillage of the cleaning solution. A tray or basket **1054** for holding jewelry items may be inserted into the bath **1052**. The basket **1054** may be provided with slots **1055** or fine openings **1056** to allow the cleaning fluid to pass but the basket **1054** is configured at the bottom **1058** to trap debris that may be dislodged from the ring **1021** or other jewelry items in the basket **1054**. The top surface **1045** of the lower housing assembly **1006** may be provided with one or more recessed portions **1057** that provide access for tabs **1060** on opposite sides of the basket **1054** for lifting the basket **1054** from the lower housing assembly **1006**. A raised member **1061** may include one or more magnets **1062**. The raised member **1061** may provide an alignment guide for placing the upper housing assembly **1003** onto the lower housing assembly **1006**. The magnets **1062** may be configured to provide sufficient force to keep the assemblies **1003**, **1006** together during use and transport, but still allow them to be manually separated by a user.

(37) Turning to FIG. **12**, the ring holder mechanism **1018** may include a ring arm motor **1063** operatively connected with a ring arm **1066** having a proximal end **1069** and a distal end **1072**. The proximal end **1069** connects with an output shaft (not visible) on the right hand side of the ring arm motor **1063** for rotational movement about an axis. The distal end **1072** of the ring holder mechanism **1018** may include a pair of opposed members **1075** and **1078** that provide for removably attaching the ring **1021** to the ring arm **1066**. The opposed members **1075** and **1078** are disposed in spaced apart relation above opening **1015**. A brush housing **1084** is visible through the opening **1015**.

(38) In FIG. **13A**, the ring arm **1066** is shown in greater detail. The opposed members **1075** and **1078** may be spring biased in opposite directions by a torsion spring. Legs extending from each end of the torsion spring may provide an outward force pushing the two opposed members **1075** and **1078** away from each other such that they are spring biased in an open position. The opposed members **1075** and **1078** may be provided with curved ends **1087** and **1090** that may have ridges defined thereon. The curved ends **1087** and **1090** may be covered with silicone sleeves. In use (best shown in FIG. **13B**), the two curved ends **1087** and **1090** are pushed together against the force of the torsion spring until they form a round shape and the ring **1021** is inserted over the curved ends **1087** and **1090** such that the opposed members **1075** and **1078** fit inside the opening in the ring **1021**. The outward force of the opposed members **1075** and **1078** against the band of the ring **1021** holds the ring **1021** in position at the end of the ring holder mechanism **1018**.

(39) From the position shown in FIG. **13A** (with the ring **1021** attached to the ring holder mechanism **1018**), the ring arm motor **1063** causes the ring arm **1066** to rotate in the clockwise direction causing the opposed members **1075** and **1078** to carry the ring **1021** through the bristles **1024**, into contact with the bristles in the brush assembly **1084** and finally into the bath **1052** in the lower housing assembly **1006**, as will be described in greater detail herein.

(40) Turning to FIGS. **16**, **17**, brush housing **1084** is disposed on the right side of the figure and will be described in detail. Brush housing **1093** is similar in configuration and therefore will not be described in detail. For clarity the brush housing **1084** is shown with the bristles removed. The

bristles on the brush housing **1084** may be stiffer than the bristles **124** at the opening **1015**. The brush housing **1084** may be comprised of a bristle mounting pad **1096** mounted on the outside and a vibration motor **1099** disposed inside of a single housing **1084**. Accordingly, the brush housing **1084** may be configured as a unitary replaceable unit that is inserted into the upper housing assembly **1003**. The brush housing may comprise a pair of projecting members **1085** and **1086** that fit into a space formed in the upper housing assembly **1003**. When the brush housing **1084** is inserted into the upper housing assembly **1003**, electrical contact with a circuit may be made by electrical connectors such as pins or the like to connect the vibration motor **1099** to the electrical circuit in the unit. The vibration motor **1099** provides a pulsating motion to the bristle mounting pad **1096** which results in pulsating bristles **1024** contacting the exposed surfaces of the ring **1021**. As shown, the bristle mounting pad **1096** may extend into the bath **1052** area below the fill line **1097** (FIG. **15**) for the cleaning solution.

(41) The bath **1052** of the lower housing assembly **1006** may be provided with a transducer **1105** mounted to the bath **1052**. The transducer **1105** may be centered and located snug against the bottom wall of the bath **1052**. The bath **1052** provides for ultrasonic cleaning in combination with the pulsating bristles.

(42) The control of the ring holder mechanism **1018** and the movement between the different positions above the bristles **1024**, through the bristles in the brush housing **1084** and into the bath **1052** may be as described above in connection with the first embodiment.

(43) A jewelry cleaning device has been described above in connection with more than one embodiment. Modifications and alterations will occur to those upon reading and understanding the preceding detailed description. For example the ring holder could be used with other cleaning mechanisms or perhaps only used with the cleaning brushes or the bath. Other modifications can also be provided. The invention, however, is not limited to only the embodiment described above. Instead, the invention is broadly defined by the appended claims and the equivalents thereof.

Claims

1. A jewelry cleaning device for cleaning a ring with a cleaning solution, the jewelry cleaning device comprising: a housing including an upper housing assembly disposed on a lower housing assembly, the upper housing assembly configured to move between an open and a closed position relative to the lower housing assembly; a bath located in the lower housing assembly and configured to hold the cleaning solution; a cleaning brush mounted to the upper housing assembly; a ring holder supported by the upper housing assembly and configured to rotate about an axis disposed on the upper housing assembly hold the ring, and a transducer cooperating with the bath for vibrating the bath wherein when the upper housing assembly is in the closed position, the ring holder being is movable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath.

2. The jewelry cleaning device of claim 1, wherein at least a portion of the cleaning brush is disposed in the cleaning solution in the bath when the upper housing assembly is in the closed position.

3. The jewelry cleaning device of claim 1, further comprising a plurality of bristles supported by the upper housing assembly and positioned above the cleaning brush, wherein the ring holder positions the ring above the plurality of bristles when in the first position and the ring holder positions the ring below the plurality of bristles when in the second position, and wherein the ring holder moves through at least a portion of the plurality of bristles.

4. The jewelry cleaning device of claim 1, wherein the cleaning brush further comprises a cleaning brush housing with bristles and a vibration motor disposed thereon.

5. The jewelry cleaning device of claim 1, wherein the ring holder includes a fixed grip and a

movable grip that is moveable with respect to the fixed grip.

6. The jewelry cleaning device of claim 1, wherein the upper housing assembly includes a lid pivotable with respect to the an upper housing between the open position and the closed position.

7. The jewelry cleaning device of claim 6, further comprising one or more magnets disposed on one of the upper housing assembly and the lower housing assembly, the magnets configured to provide a force holding the upper housing assembly and the lower housing assembly together.

8. The jewelry cleaning device of claim 6, further comprising a raised portion on one of the upper housing assembly and lower housing assembly and an opening on the other of the upper housing assembly and the lower housing assembly such that the raised portion provides an alignment guide for connecting the upper housing assembly to the lower housing assembly.

9. The jewelry cleaning device of claim 1, wherein the ring holder comprises a pair of opposed members spring biased in the open position, the opposed members configured to fit inside a band portion of the ring to hold the ring in position.

10. The jewelry cleaning device of claim 9, further comprising a silicone material disposed on at least one of the opposed members.

11. A jewelry cleaning device for cleaning a ring with a cleaning solution, the jewelry cleaning device, comprising: a housing; a bath located in the housing and configured to hold the cleaning solution; a cleaning brush housing mounted to the housing, the cleaning brush housing having bristles and a vibration motor disposed thereon; a ring holder connected with the housing and configured to hold the ring, the ring holder being movable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath; a transducer cooperating with the bath for vibrating the bath; wherein the housing further comprises an upper housing assembly removably attached to a lower housing assembly; wherein the bath is exposed and removable from the lower housing assembly when the upper housing assembly is in the open position with respect to the lower housing assembly; and, further comprising a first electrical connector electrically connected with the transducer, and a second electrical connector for connecting with a power source, wherein the first electrical connector is disconnectable with the second electrical connector when the bath is removed from the lower housing assembly.

12. A jewelry cleaning device for cleaning a ring with a cleaning solution, the jewelry cleaning device comprising: a housing including an upper housing assembly disposed on a lower housing assembly, the upper housing assembly configured to move between an open and a closed position relative to the lower housing assembly; a bath located in the lower housing assembly and configured to hold the cleaning solution; a cleaning brush mounted to the upper housing assembly, the cleaning brush having a cleaning brush housing with a vibration motor disposed thereon; a ring holder supported by the upper housing assembly and configured to rotate about an axis disposed on the upper housing assembly, wherein when the upper housing assembly is in the closed position, the ring holder being is movable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath; wherein when the ring holder moves from the first position to the second position the ring passes through a third position where it engages with the cleaning brush; the ring holder having a pair of opposed members spring biased in an open position, the opposed members configured to rotate toward each other against the force of the spring to fit inside a band portion of the ring to hold the ring in position; and, a transducer cooperating with the bath for vibrating the bath.

13. A jewelry cleaning device for cleaning a ring with a cleaning solution, the jewelry cleaning device comprising: a housing; a bath located in the housing and configured to hold the cleaning solution; a cleaning brush mounted to the housing, the cleaning brush having a cleaning brush housing with bristles and a vibration motor disposed thereon; a ring holder connected with the housing and configured to hold the ring, the ring holder being moveable with respect to the housing

between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath, the ring holder having a pair of opposed members spring biased in an open position, the opposed members configured to rotate toward each other against the force of the spring to fit inside a band portion of the ring to hold the ring in position; a transducer cooperating with the bath for vibrating the bath; and, wherein the housing further comprises an upper housing assembly and a lower housing assembly with one or more magnets disposed on at least one of the top surface of the lower housing assembly and the bottom surface of the upper housing assembly.

14. A jewelry cleaning device for cleaning a ring with a cleaning solution, the jewelry cleaning device comprising: a housing; a bath located in the housing and configured to hold the cleaning solution; a cleaning brush mounted to the housing, the cleaning brush having a cleaning brush housing with bristles and a vibration motor disposed thereon; a ring holder connected with the housing and configured to hold the ring, the ring holder being moveable with respect to the housing between a first position where the ring holder positions the ring offset from the bath and the cleaning brush and a second position where the ring holder positions the ring in the bath, the ring holder having a pair of opposed members spring biased in an open position, the opposed members configured to rotate toward each other against the force of the spring to fit inside a band portion of the ring to hold the ring in position; a transducer cooperating with the bath for vibrating the bath; and, wherein at least a portion of the cleaning brush is disposed in the cleaning solution in the bath.
